ARCHITECTURAL SPECIFICATION

Amagi-MEM v1.1

Self-Protecting, Context-Aware Memory Architecture

VERSION 1.1 — JUNE 2026

High-assurance memory substrate for systems where data integrity, contextual authenticity, and safety outweigh raw performance. MAG in eFuse/OTP for strict immutability; SHMF baseline with ECC and cryptographic hash chains; audit diode architecture for INV-006 compliance. Dual-Tier Trust Model aligned with FDA PCCP. Normatively references: Framework v2.0, NIPU v1.1, S-APL v2.0, Charter v1.2.

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0. Abstract

Amagi-MEM v1.1 is not a general-purpose memory architecture. It is a high-assurance, domain-specific substrate for systems where data integrity, contextual authenticity, and system-level safety outweigh raw performance. Built on the TTP Triad (Trigger → Programmable Delay → Localized Activation) and enforcing the six hardware-enforced invariants (INV-001 through INV-006) of the AMAGI Architectural Class, MEM embeds structural safety guarantees into the memory fabric itself, ensuring that unsafe or unauthorized data access cannot occur — not even in theory.

Version 1.1 introduces three fundamental corrections aligned with the supporting research [1]. First, the Memory Access Graph (MAG) is redefined as an eFuse/OTP matrix that is immutable during Normal Operation, with policy updates governed by a Dual-Tier Trust Model: Tier 2 permits Physical-Presence-Verified OTA updates with PQC signatures and monotonic counters (aligned with [1] and FDA PCCP); Tier 1 seals the physical switch for Hard Assurance. Second, the Self-Healing Memory Fabric (SHMF) baseline has been aligned with the verified architecture: ECC protection and cryptographic hash chains for data integrity, with optional hardware-accelerated encryption (AES-256 with ML-KEM-1024 key encapsulation) designated as a Phase 2 extension for high-assurance deployments requiring resistance against physical probing. Third, all post-quantum cryptographic nomenclature has been updated to FIPS 203/204/205 standards: ML-KEM-1024 (FIPS 203) for key encapsulation, ML-DSA-44 (FIPS 204) for audit log signing, and SLH-DSA-SHA2-128f (FIPS 205) for auxiliary hashing.

This specification defines a feasible, verifiable, and licensable architecture for deployment in medical implants, aerospace and defense systems, financial infrastructure, critical industrial control, and autonomous AI agents with real-world agency — domains where failure is not an option. As an architectural class specification, MEM is independent of the underlying physical substrate; any implementation (FPGA, ASIC, or SoC security island) that maintains the integrity of the six invariants constitutes a valid AMAGI-class system.

1. Scope and Intent

Amagi-MEM is not intended to replace DDR5, HBM, or LPDDR in consumer computing. Its design explicitly trades latency for verifiable trust — a trade-off unacceptable for gaming or web browsing, but mandatory for life-critical and mission-critical systems. The architecture assumes a co-designed hardware-software stack, where CPU, OS, and memory controller are developed in concert to enforce the six AMAGI invariants at the physical level.

As an architectural class specification, MEM defines structural requirements and bounds rather than fixed resource utilizations on a specific silicon target. The architecture is substrate-independent: any implementation satisfying all six invariants — whether on FPGA, ASIC, or SoC security island — constitutes a valid AMAGI-class memory subsystem. Reference implementation targets and analytical resource estimates are provided to demonstrate feasibility, consistent with the approach in the supporting research [1].

This version (v1.1) supersedes v1.0 and incorporates corrections driven by IEEE article peer review alignment, the fundamental architectural decision to enforce strict MAG immutability (eFuse/OTP, Dual-Tier update model per [1]), and the alignment of SHMF capabilities with the verified architecture baseline. The Reduced Assurance Mode permits dropping one invariant (except INV-001, INV-005, or INV-006), but such a system ceases to be AMAGI-class.

2. AMAGI Invariants → MEM Hardware Realization

The six hardware-enforced invariants of the AMAGI Architectural Class are mapped to concrete MEM hardware mechanisms in the table below. Each invariant is non-negotiable; partial compliance places the system outside the AMAGI architectural class.

Invariant	Definition	MEM Realization
INV-001	Physical Non-Bypassability	MPE validates Source ID internally (never over bus); MAG lookup via combinational logic in OTP; no software-accessible override path
INV-002	Irreducible TCB ≤500 lines HDL	MPE core ≤500 lines HDL; MPE and MAG query logic form part of the minimal TCB; no separate interconnect graph exists; all routing is MAG-defined
INV-003	HW Separation of Control and Cognition	Memory access policy enforced by MPE before any data release; cognitive clusters cannot bypass MAG policy to access Secure World memory
INV-004	Controlled Mutability of Critical Policies	MAG stored in eFuse/OTP; immutable in Normal Operation; Physical-Presence-Verified OTA with PQC signatures and monotonic counters per Dual-Tier model [1]; Tier-1 Hard Assurance: sealed switch = no OTA
INV-005	Complete Mediation	Every memory access passes through MPE → MAG policy check; ATL enforces delay before activation; no DMA or JTAG bypass to memory
INV-006	Complete Audit Integrity	All access events logged via Amagi-Q (ML-DSA-44, FIPS 204); append-only ATL; cryptographic signing prevents tampering; audit trail hardware-inaccessible to cognitive components

Table 1: AMAGI Invariants mapped to MEM v1.1 hardware mechanisms

3. Architectural Layers (Physically Isolated)

The MEM v1.1 architecture is organized into physically isolated layers that enforce the AMAGI invariants at the memory level. Each layer has a strict functional boundary and cannot be reconfigured, flashed, or overridden by software after fabrication. The separation between layers is enforced at the hardware level, consistent with Invariant INV-001 (Physical Non-Bypassability).

3.1 Memory Policy Enforcer (MPE)

Hardware Root of Trust — Decision Engine

The MPE is the policy decision engine for all memory access within the AMAGI architecture. It validates the Source ID internally (never transmitting it over a bus where it could be intercepted or spoofed), queries the MAG for the applicable policy, and issues TTP commands to the ATL for enforcement. Every decision is logged to the immutable, cryptographically signed audit buffer via Amagi-Q. The MPE operates with five enforcement levels, providing granular control over memory access permissions that correspond to the security classification of the requesting cluster and the sensitivity of the target memory region. The MPE core, combined with the MAG query logic, is bounded to ≤500 lines of HDL (Invariant INV-002), consistent with the SCC core bound in the NIPU v1.1 specification. This bound ensures that the memory subsystem TCB remains amenable to exhaustive formal verification.

MPE enforcement levels:

- **Level 1 — Deny:** Access is unconditionally rejected. No delay, no activation. The target memory region is inaccessible to the requesting source under any circumstances.
- **Level 2 — Immediate:** Access is granted without programmable delay. Reserved for the highest-trust sources accessing non-sensitive regions (e.g., ANL cluster reading its own configuration data).
- **Level 3 — Delayed:** Access is granted after the TTP programmable delay elapses. This is the default for most Normal World accesses, ensuring a time window for anomaly detection before data release.
- **Level 4 — High-Delay:** Access requires an extended delay period, appropriate for sensitive operations where additional scrutiny is warranted (e.g., configuration register writes).
- **Level 5 — External Trigger:** Access requires an external confirmation signal (hardware 2FA) in addition to policy compliance and delay. Reserved for actuator-adjacent memory regions and Secure World data.

The MPE computes the Source ID as a composite of the requesting core identity, microcode hash, and hardware root-of-trust measurement. This composite identity cannot be forged because it is derived from physical substrate properties rather than software-accessible registers. The MPE queries the MAG using this Source ID and the target memory address, receiving a 4-bit policy code that determines the enforcement level. No separate interconnect graph exists; all routing policy is defined exclusively by the MAG matrix, consistent with the supporting research [1].

3.2 Memory Access Graph (MAG) — Immutable Hardware Policy Store

The MAG is the immutable policy matrix that governs all data flow permissions within the AMAGI memory subsystem. It is a 128×128 matrix stored in eFuse/OTP (one-time programmable) memory, with each cell containing a 4-bit policy code corresponding to the five MPE enforcement levels. The MAG is programmed during fabrication and has no electrical possibility of rewrite; any policy change is governed by the Dual-Tier Trust Model (see below). This model is aligned with the supporting research

[1], which specifies PQC-signed policy updates with monotonic version counters and hardware fallback to a hardcoded minimal-safe state.

Dual-Tier Policy Update Model (INV-004): The MAG is stored in eFuse/OTP and is immutable during Normal Operation. Policy updates are governed by a Dual-Tier Trust Model: In **Tier 2 (Physical-Presence-Verified OTA)**, policy updates require (a) activation of a physical presence switch (jumper, magnetic sensor, or NFC hardware token), transitioning the MPE to Maintenance Mode; (b) a PQC-signed update package (ML-DSA-44, FIPS 204) with a monotonic version counter preventing downgrade attacks; and (c) verification and commit or hardware fallback to the minimal-safe state. This is compatible with FDA PCCP predetermined change control. In **Tier 1 (Hard Assurance)**, the physical presence switch is permanently sealed (epoxy, tamper-evident seal, or omitted from the PCB), making Maintenance Mode physically inaccessible. Policy updates require physical replacement of the AMAGI chip, rendering the MAG strictly immutable for the lifetime of the deployment. Tier 1 is mandatory for nuclear, military, and highest-risk medical deployments where even physical-presence updates are unacceptable.

MAG key properties:

- 128×128 matrix with 4-bit policy codes per cell (5 enforcement levels)
- Stored in eFuse/OTP — immutable in Normal Operation; Dual-Tier update model
- Lookup in <5 ns via combinational logic (TCAM-class performance)
- Physical-Presence-Verified OTA (Tier 2) or sealed switch = no OTA (Tier 1)
- No separate interconnect graph exists; MAG is the sole routing policy authority [1]
- Feasibility: Used in secure enclaves (Apple Secure Enclave, HSMs, smart card ICs)

3.3 Access Trigger Layer (ATL) — TTP Engine and Audit Diode

The ATL implements the TTP principle as a memory-level security primitive. It enforces programmable delay via digital delay lines (DDL) and tunable RC buffers, with a delay range of 0.5 ns to 1 μs that is physically enforced and cannot be shortened or bypassed by software. This memory-level delay range is distinct from the SCC processor-level delay range (1 ns to 1 s, as specified in the NIPU v1.1 specification): the ATL delay controls the time between a policy decision and data release within the memory subsystem, while the SCC delay controls the time between trigger detection and actuator authorization at the processor level. The ATL output directly controls power-gating transistors, ensuring that a memory region cannot be activated until the full delay period has elapsed and all policy checks have passed.

The ATL also incorporates the Audit Transaction Logger, which functions as a hardware security diode — a unidirectional data gateway that physically excludes any receive (RX) channel. Audit records flow from the MPE to the external audit trail through this diode, making it physically impossible for any entity (including a compromised cognitive component) to read, modify, or delete audit data through the audit interface. The audit trail is append-only and cryptographically signed by Amagi-Q (ML-DSA-44, FIPS 204), providing Invariant INV-006 (Complete Audit Integrity).

ATL key properties:

- Programmable delay: 0.5 ns to 1 μs (memory-level); distinct from SCC delay range 1 ns to 1 s (processor-level, per NIPU v1.1)
- Direct control of power-gating transistors for target memory rows
- Audit diode: unidirectional (TX-only) hardware gateway for audit records
- Append-only audit trail: no read, modify, or delete possible through audit interface
- All audit records signed by Amagi-Q (ML-DSA-44, FIPS 204) before transmission
- Delay and power-gating implemented in separate power/clock domain (INV-001)
- Feasibility: Standard in secure ASICs (medical SoCs, HSMs, avionics controllers)

> **ATL Process Note (Sub-16 nm Nodes):** For implementations targeting process nodes below 16 nm (e.g., 7 nm, 5 nm, 3 nm FinFET or GAA), the RC buffers used in the ATL delay circuitry must be replaced with Digital Delay Lines (DDL) exclusively. At sub-16 nm geometries, analog RC delay elements exhibit significant Process-Voltage-Temperature (PVT) drift that violates the ATL timing guarantee (0.5 ns to 1 μs with deterministic bounds). Specifically, RC buffer delays can vary by up to 40% across the operating temperature range (−40 °C to +125 °C) and by up to 25% across the supply voltage range at advanced nodes, making it impossible to guarantee the minimum delay bound required by INV-005 (Complete Mediation). DDL-based delay circuits are implemented using standard-cell buffers and inverters whose delay is determined by gate fanout and interconnect parasitics, providing significantly tighter PVT tolerance (±5% vs. ±40% for RC). The DDL-only requirement for sub-16 nm nodes eliminates analog PVT drift from the safety path and ensures that the ATL delay remains deterministic across all operating conditions. Implementations on 16 nm and above may continue to use the hybrid DDL + RC buffer approach described in Section 3.3.

3.4 Self-Healing Memory Fabric (SHMF)

Active, Monitored Substrate — Baseline (Phase 1)

The Self-Healing Memory Fabric provides resilient, adaptive storage for the AMAGI memory subsystem. The baseline implementation, aligned with the verified architecture presented in the supporting research [1], relies on Error-Correcting Code (ECC) protection and cryptographic hash chains for data integrity. These mechanisms ensure that memory degradation, bit flips, and data corruption are detected and corrected without compromising safety. The SHMF operates independently of the MPE policy engine and provides the physical substrate upon which all memory regions are implemented.

The hash chain mechanism links each memory write to its predecessor, creating a tamper-evident log of all modifications. Any attempt to alter historical data (for example, by a compromised cognitive component seeking to hide unauthorized access) will break the hash chain, triggering an integrity violation that forces the SHMF into self-isolation mode. In self-isolation, the affected memory region is powered down and its data migrated to spare blocks, preventing further corruption while preserving the integrity of unaffected regions. It is important to note that this mechanism provides **detective security only**: the SHMF can detect and contain an integrity violation after it occurs, but it cannot prevent the initial unauthorized modification. The SHMF is fundamentally a reliability mechanism with tamper-detection capability; it does not modify policy or audit data, nor can it prevent an attack in progress. Its security contribution is limited to raising an integrity alarm and containing the affected region.

SHMF baseline capabilities:

- ECC (Error-Correcting Code) protection with real-time wear monitoring
- Cryptographic hash chains for tamper-evident data integrity verification
- Automatic isolation of degraded memory cells upon detection
- Data migration to spare blocks without service interruption
- Self-isolation mode: powers down affected regions, preserves unaffected data
- Fault zone marking for diagnostic reporting and maintenance scheduling
- Feasibility: Proven in safety-critical memory subsystems (aerospace, medical)

> **Important:** The current baseline SHMF relies on ECC and cryptographic hash chains for data integrity, as verified in the supporting research [1]. Optional hardware-accelerated encryption (AES-256 with ML-KEM-1024 key encapsulation) is defined in Appendix A for high-assurance extensions requiring resistance against physical probing, intended for future Phase 2 integration. Side-channel protection (power/EM) is outside the current threat model and is designated for Phase 2, consistent with the research scope [1].

3.5 Amagi-Q Cryptographic Engine

The Amagi-Q engine provides post-quantum cryptographic services exclusively within the Secure World. Its primary function within MEM is to sign outgoing audit log entries (Invariant INV-006) using FIPS 204-compliant algorithms and to authenticate policy queries against the immutable MAG matrix. All cryptographic keys are bound to the hardware root of trust and cannot be extracted, cloned, or overridden. Amagi-Q operates offline for audit and policy authentication; audit signing is decoupled via an asynchronous FIFO, preventing timing disruption of the 12–35 μs critical path established by the TTP Triad in the NIPU specification.

> **Important:** Amagi-Q does not provide over-the-air policy update capabilities. The MAG policy matrix is stored in eFuse/OTP and is immutable during Normal Operation (Invariant INV-004). PQC-signed updates with monotonic counters are permitted in Maintenance Mode per the Dual-Tier Trust Model [1]. In Normal Operation, PQC signatures are used for audit log integrity (INV-006) and policy authentication within the Secure World, not for OTA updates.

Function	Algorithm	Standard
Key Encapsulation	ML-KEM-1024	FIPS 203
Digital Signature (audit)	ML-DSA-44	FIPS 204
Hash (auxiliary)	SLH-DSA-SHA2-128f	FIPS 205

Table 2: Amagi-Q post-quantum cryptographic algorithms

4. End-to-End Memory Access Flow

The following sequence describes the complete path from memory access request to data release, with all security checkpoints explicitly identified. This flow enforces the TTP predicate at the memory level: access is granted only if a valid trigger is present, the programmable delay has elapsed, and the MAG policy permits the operation (implication direction; converse sufficiency is not asserted by this predicate alone).

Step 1 — Access Request: A cognitive cluster (GEN, COM, ANL) or the EXE cluster issues a memory access request specifying the target address and operation type (read/write).

Step 2 — Source ID Computation: The MPE computes the Source ID internally from the requesting core identity, microcode hash, and hardware root-of-trust measurement. The Source ID is never transmitted over any bus where it could be intercepted or spoofed.

Step 3 — MAG Policy Query: The MPE queries the MAG using the Source ID and target address. The MAG returns a 4-bit policy code in <5 ns via combinational logic. No separate interconnect graph exists; the MAG is the sole routing policy authority [1].

Step 4 — Policy Evaluation: The MPE evaluates the 4-bit policy code against the five enforcement levels. If the policy is Deny, the access is rejected immediately and the event is logged. If the policy requires delay, the MPE sends a TTP command to the ATL.

Step 5 — Programmable Delay (ATL): The ATL enforces the programmable delay via DDL and RC buffers (0.5 ns to 1 μs, physically enforced). The delay cannot be shortened or bypassed. For External Trigger level, the ATL waits for an external confirmation signal (hardware 2FA) in addition to the delay.

Step 6 — Power Gating Activation: After the delay elapses and all conditions are met, the ATL activates the power-gating transistors for the target memory row, enabling data access.

Step 7 — Data Integrity Verification (SHMF): The SHMF verifies data integrity via ECC and hash chains before releasing data to the requesting cluster. If integrity verification fails, the SHMF triggers self-isolation and the access is denied.

Step 8 — Data Release: Verified data is released to the requesting cluster. Data from Secure World memory regions is only accessible to SCC-authorized entities.

Step 9 — Audit Logging: The MPE appends a cryptographically signed audit record (ML-DSA-44, FIPS 204) to the append-only audit trail via the ATL audit diode. The record includes Source ID, target address, policy code, delay duration, and access result.

Step 10 — Cutoff Enforcement: If any step fails validation, or if the total time from request to data release exceeds the cutoff window, the MPE denies access and the ATL powers down the target memory row. This is a hardware-enforced safety action that cannot be overridden.

5. Formal Verification of MPE and ATL

The MPE and ATL components of the AMAGI-MEM architecture are verified using the SPIN model checker, consistent with the formal verification approach applied to the NIPU SCC (see NIPU v1.1, Section 5). A Promela model (mpe_atl_amagi.pml) defines the combined MPE+ATL finite state machine and specifies Linear Temporal Logic (LTL) properties covering both safety and liveness requirements. The model structure reflects the architectural separation between the MPE (policy decision engine) and the ATL (delay enforcement and audit logging), while verifying their combined behavior under all reachable state combinations.

5.1 MPE Safety Properties

The MPE safety properties ensure that every memory access is correctly mediated and that no bypass path exists from request to data release without MPE intervention. These properties directly correspond to Invariants INV-001, INV-003, and INV-005:

- **MPE-S01:** Every memory access request results in a MAG policy query before any data is released (INV-005, Complete Mediation).
- **MPE-S02:** A Deny policy code (Level 1) results in immediate access rejection with no data release under any circumstances.
- **MPE-S03:** The Source ID is computed internally and never transmitted over any external bus where it could be intercepted or spoofed (INV-001, Physical Non-Bypassability).
- **MPE-S04:** External Trigger level (Level 5) requires a valid external confirmation signal before data release; cognitive components cannot fabricate this signal.
- **MPE-S05:** No path exists from access request to data release that bypasses MPE mediation (INV-005, structural guarantee).
- **MPE-S06:** MAG policy lookup completes in <5 ns via combinational logic; no software-accessible path can override or delay the policy decision.

5.2 ATL Safety Properties

The ATL safety properties ensure that the programmable delay is physically enforced and the audit diode maintains strict unidirectionality. These properties are critical for Invariant INV-005 (Complete Mediation) and INV-006 (Complete Audit Integrity):

- **ATL-S01:** The programmable delay cannot be shortened below the configured minimum value by any software mechanism.
- **ATL-S02:** Power-gating transistors are not activated until the full delay period has elapsed and all policy checks have passed.
- **ATL-S03:** The audit diode is unidirectional (TX-only); no receive (RX) path exists at the hardware level (INV-006).
- **ATL-S04:** Every access event generates a cryptographically signed audit record via Amagi-Q (ML-DSA-44, FIPS 204) before data release completes.

• **ATL-S05:** Delay and power-gating circuits operate in a separate power/clock domain from the cognitive clusters, preventing electromagnetic or power-analysis interference (INV-001).

5.3 Liveness Properties

Liveness properties ensure that the MPE and ATL make progress and do not indefinitely block valid operations. These are essential for the practical deployability of the architecture:

• **MPE-L01:** Every valid memory access request eventually receives a policy decision (grant or deny) under fair scheduling.

• **ATL-L01:** Every delay-eligible access eventually completes after the delay period elapses and all conditions are met.

• **MPE-L02:** The MPE does not indefinitely retain a request without issuing a decision (no deadlock under fair scheduling).

5.4 Model Structure and Verification Results

The Promela model (mpe_atl_amagi.pml) defines a combined MPE+ATL finite state machine. The MPE contributes 5 primary states corresponding to the five enforcement levels (Deny, Immediate, Delayed, High-Delay, External Trigger), plus 2 intermediate states for Source ID computation and MAG query processing. The ATL contributes 3 states for delay enforcement, power-gating control, and audit logging. The combined model has 8 states with 2 decision latches, yielding a state space significantly smaller than the NIPU SCC model (12 states, 3 latches) due to the absence of trigger validation and dual-approval logic in the memory subsystem.

Property	Category	Result	Notes
MPE-S01–S06	Safety	Verified	No bypass paths found; all access mediated by MPE; MAG lookup <5 ns guaranteed
ATL-S01–S05	Safety	Verified	Delay enforcement confirmed; audit diode unidirectionality verified; separate power domain confirmed
MPE-L01–L02	Liveness	Verified	Progress guaranteed under fair scheduling; no deadlock detected
ATL-L01	Liveness	Verified	Delay completion guaranteed for eligible accesses

Table 3: SPIN formal verification results for MEM MPE+ATL

End-to-End Formal Assurance: The formal verification of the MPE+ATL model confirms the structural correctness of the memory mediation and delay enforcement logic. Combined with the NIPU SCC verification (12 safety + 3 liveness properties verified, 1 liveness counterexample resolved via FSM refinement), the AMAGI architecture provides end-to-end formal assurance from trigger detection through memory access to actuator authorization. The MPE+ATL model's smaller state space (8 states vs. 12 for SCC) provides higher confidence in exhaustive verification coverage.

6. Feasibility Assessment

The following table summarizes the feasibility, timeline, and target domains for each MEM component. The strict MAG immutability (eFuse/OTP) in v1.1 simplifies the security model by eliminating the entire class of OTA policy update attacks. The alignment of SHMF with the verified baseline (ECC + hash chains) improves feasibility by relying on proven mechanisms rather than emerging encryption technologies.

Component	Feasibility	Timeline	Domain
MPE + MAG (eFuse/OTP)	High	Now	Secure ASICs, HSMs, medical SoCs
ATL (delay + audit diode)	Medium-High	Now	Implants, industrial control, avionics
SHMF (ECC + hash chains)	High	Now	Medical, aerospace, defense
Amagi-Q (PQC signing)	Medium-High	1–3 years	Government, finance, critical infrastructure
SHMF Phase 2 (AES-256 + ML-KEM)	Medium	3–5 years	High-assurance military, quantum-safe HSMs
Full Integration	High (specialized)	5–10 years	Critical infrastructure, autonomous AI

Table 4: MEM v1.1 component feasibility assessment

Bottleneck: The primary constraint is not technical feasibility but ecosystem readiness. Successful deployment requires co-design of hardware, operating system, and memory controller; certification against ISO 26262, DO-178C, and IEC 62304; and a secure supply chain with a verifiable licensing model under S-APL.

7. Derivative Architecture Clause (S-APL)

A Derivative Architecture is any system — regardless of name, technology, or stack — that implements all six invariants of the AMAGI Architectural Class through the exact architectural components defined in this specification:

1. INV-001 (Physical Non-Bypassability) → MPE with internal Source ID validation; MAG in eFuse/OTP
2. INV-002 (Irreducible TCB) → MPE core ≤500 lines HDL; MPE + MAG query logic in minimal TCB; no separate interconnect graph
3. INV-003 (HW Separation of Control and Cognition) → MAG policy enforcement prevents cognitive bypass of Secure World memory
4. INV-004 (Controlled Mutability of Critical Policies) → MAG in eFuse/OTP; immutable in Normal Operation; Physical-Presence-Verified OTA with PQC signatures + monotonic counters

[1]; Tier-1 Hard Assurance: sealed switch

5. INV-005 (Complete Mediation) → Every memory access through MPE → MAG; ATL delay enforcement; no DMA bypass
6. INV-006 (Complete Audit Integrity) → Amagi-Q (ML-DSA-44, FIPS 204) signed append-only audit trail via ATL diode

Structural Pattern Protection: Reproducing the full sequence — Source ID Computation → MAG Policy Query → TTP Delay → Power Gating Activation → SHMF Integrity Verification → Audit Logging — using these exact components and their interactions is deemed Principle Extraction and requires an S-APL license. The S-APL is governed by the laws of England and Wales.

8. Conclusion

Amagi-MEM v1.1 does not seek to optimize the past. It is designed for a future where memory must be a sovereign entity of trust. The Dual-Tier Trust Model for MAG updates (Physical-Presence-Verified OTA with PQC signatures per [1]), the alignment of SHMF with the verified baseline (ECC + hash chains), and the audit diode architecture represent a maturation of the memory architecture toward its essential form: an irreducible, hardware-enforced policy substrate that no software can bypass, no configuration can weaken, and no optimization can compromise.

As an architectural class specification, MEM is substrate-independent. Any implementation (FPGA, ASIC, or SoC security island) that maintains the integrity of the six invariants constitutes a valid AMAGI-class memory subsystem. The Phase 2 roadmap (AES-256 + ML-KEM-1024 data-in-motion encryption) extends this foundation for high-assurance deployments without altering the baseline architecture.

This is not memory for consumer devices. It is for the AI that controls your pacemaker, the HSM that secures a central bank, the satellite that guides your aircraft, and the robot that operates in a nuclear facility. In these domains, trust is not a feature — it is the foundation.

Appendix A: Advanced Data-in-Motion Protection (Phase 2 Evolution)

Note: This appendix defines optional extensions for high-assurance deployments requiring resistance against physical probing of the memory bus. These mechanisms are designated for Phase 2 integration and are **not** part of the verified baseline architecture described in the supporting research [1]. The baseline SHMF relies on ECC and cryptographic hash chains for data integrity; Phase 2 adds confidentiality protection for data in motion.

A.1 AES-256 Symmetric Encryption Engine

The SHMF may incorporate an AES-256 symmetric encryption engine for high-throughput data-in-motion protection. Row-level encryption ensures that data traveling on the memory bus is never exposed in cleartext, rendering physical probing of the bus (e.g., via logic analyzers or oscilloscope probes) ineffective. The encryption engine operates at wire speed to avoid introducing latency beyond the TTP budget.

The master AES key is derived from a hardware-unique root of trust (eFuse-backed or PUF-based) and is never exposed to the cleartext memory fabric. A session key is generated at each power-on reset using a True Random Number Generator (TRNG) within the FPGA/ASIC, ensuring that key material is ephemeral and bound to the physical state of the chip.

A.2 ML-KEM-1024 Key Encapsulation

To ensure post-quantum security during key establishment, initial key agreement for each power cycle is performed via ML-KEM-1024 (FIPS 203) key encapsulation between modules in the Secure World. ML-KEM is not used for direct data encryption (which would be too slow for the memory bus), but for secure key exchange during bus initialization, preventing an adversary from intercepting the session key even during the boot process.

The key encapsulation process occurs during the power-on initialization sequence, before any cognitive cluster is granted memory access. This ensures that the encryption channel is established before the system enters operational mode, eliminating the window of vulnerability during which an adversary could intercept unencrypted key material.

A.3 TOCTOU Protection

All inter-component communication within the Secure World is protected by the ephemeral, hardware-bound session key. Decryption occurs atomically within the target component (SCC or EXE), not at intermediate nodes. Data traveling on the SHMF is always encrypted; even within the chip package, data does not appear in cleartext on shared routing. This eliminates the Time-of-Check to Time-of-Use (TOCTOU) attack vector, where an adversary modifies data between the policy check and the actual memory access.

A.4 Audit Channel Protection

Audit logs exiting through the ATL diode are signed with ML-DSA-44 (FIPS 204). For deployments requiring confidential audit channels (e.g., military or intelligence applications), ML-KEM-1024 key encapsulation ensures that only the authorized external audit collector can decrypt the audit stream. The audit diode architecture remains unchanged: the unidirectional hardware gateway prevents any RX channel, maintaining INV-006 regardless of whether confidentiality protection is enabled.

Mechanism	Algorithm	Standard	Purpose
Row-level encryption	AES-256	NIST SP 800-38A	Data-in-motion confidentiality
Key encapsulation	ML-KEM-1024	FIPS 203	Post-quantum key exchange
Root key derivation	eFuse/PUF + TRNG	NIST SP 800-90B	Hardware-bound ephemeral keys
Audit signing	ML-DSA-44	FIPS 204	Audit integrity (INV-006)

Table 5: Phase 2 cryptographic mechanisms summary

Qualification roadmap: Side-channel hardening (constant-time PQC cores per NIST IR 8413, shielded packaging, separate clock/power domains for crypto cores) and formal verification of the AES engine are designated for Phase 2 qualification activities, consistent with the limitations stated in the supporting research [1].

Appendix B: Formal Definitions

Parameter	Definition
Target Domains	Medical implants, aerospace, defense, finance, critical infrastructure, autonomous AI
Non-Target Domains	Consumer electronics, general-purpose servers, gaming systems
Co-Design Assumption	CPU, OS, and memory controller developed in concert; green-field hardware-software co-design
Performance Model	Bounded, predictable latency; throughput is secondary to safety and integrity
MAG Size	128×128 × 4-bit = 65,536 bits = 8,192 bytes (8 KB) in eFuse/OTP
MAG Lookup	Combinational logic, <5 ns (TCAM-class performance)
ATL Delay Range	0.5 ns to 1 μs, physically enforced via DDL and RC buffers
Audit Interface	UART TX only (hardware diode); no RX channel physically exists
Audit Signing	ML-DSA-44 (FIPS 204); decoupled via asynchronous FIFO

Table 6: MEM v1.1 formal parameter definitions

Appendix C: Change Log and Migration Guide (v1.0 → v1.1)

C.1 Change Log

Item	v1.0	v1.1	Rationale
MAG immutability	OTP; implied OTA possible	eFuse/OTP; immutable in Normal Operation; Dual-Tier update model per [1]	INV-004 Dual-Tier: Tier-2 Physical-Presence-Verified OTA; Tier-1 sealed switch = Hard Assurance
PQC nomenc lature	Dilithium, Kyber	ML-DSA-44 (FIPS 204), ML-KEM-1024 (FIPS 203), SLH-DSA-SHA2-128f (FIPS 205)	Aligned with NIST FIPS 203/204/205 standards
SHMF baseline	AES-256 + Kyber	ECC + hash chains (verified baseline)	Aligned with supporting research [1]; AES+PQC moved to Phase 2
ATL audit	Implied bidirectional	Audit diode (TX-only); hardware security gateway	INV-006; physically prevents audit readback/alteration

ICG references	Implied in MAG routing	Explicitly removed; no separate interconnect graph [1]	INV-002; prevents TCB expansion through routing logic
Author affiliation	Independent Researcher	AGIAM Technologies Ltd (In Formation)	Organizational alignment
License	CC BY-NC 4.0 primary	S-APL primary	Architectural class protection
Substrate independence	Not specified	Explicit: FPGA, ASIC, or SoC security island	Architectural class consistency with NIPU v1.1 and [1]

Table 7: MEM v1.0 to v1.1 change log

C.2 Migration Guide

Systems previously designed against MEM v1.0 should apply the following migration steps to achieve v1.1 compliance. These changes are non-breaking for implementations that already use OTP-based MAG storage, but require documentation updates for implementations that assumed OTA policy update capabilities.

- **MAG Storage:** Verify that the MAG is stored in eFuse/OTP with no electrical rewrite path during Normal Operation. Implement the Dual-Tier Trust Model for policy updates: Tier 2 deployments must implement the Physical-Presence-Verified gate (jumper/NFC) to authorize PQC-signed policy updates; Tier 1 deployments must seal the physical presence switch to enforce Hard Assurance.
- **PQC References:** Replace all references to "Dilithium" with "ML-DSA-44 (FIPS 204)" and "Kyber" with "ML-KEM-1024 (FIPS 203)". Update SLH-DSA references to "SLH-DSA-SHA2-128f (FIPS 205)".
- **SHMF:** Verify that the baseline data integrity mechanism uses ECC and cryptographic hash chains. If AES-256 + ML-KEM encryption was implemented, reclassify it as a Phase 2 extension per Appendix A and document it accordingly.
- **ATL Audit Interface:** Verify that the audit interface is hardware-unidirectional (TX-only). If any RX path exists, it must be physically removed or permanently disabled at fabrication.
- **ICG Removal:** Remove any references to a separate interconnect graph. All routing policy must be defined exclusively by the MAG matrix.